

# ADARSH SINGH

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## PROFESSIONAL SUMMARY

Backend Engineer with 2 years of production experience designing distributed, high-concurrency systems — payment and credit infrastructure, real-time booking engines, and AI-powered media pipelines — on AWS. Core strengths in Node.js, NestJS, TypeScript, Golang, PostgreSQL, Redis, BullMQ, Kafka, and gRPC, with hands-on expertise in Two-Phase Commit, distributed locking, idempotency, and exactly-once job processing. Delivered systems that cut search latency by 40%, reduced indexing lag by over 90%, and cut UI lag by 70% in production.

## PROFESSIONAL EXPERIENCE

### Backend Engineer — Node.js / NestJS / TypeScript, Matchbest Software Pvt Ltd

Jan 2026 – Present

Product: xelta.ai (Live SaaS AI Platform) — Gurugram, India

#### Project: PixelCut — Browser-Based AI Video Editor

- Optimized a browser-based video editor (Next.js, React) by offloading video decoding, rendering, and audio processing to dedicated Web Workers with OffscreenCanvas, cutting UI lag by ~70% and enabling smooth editing on low-end laptops.
- Architected a distributed job-processing backend of 4 microservice workers (media processing, captioning, export, AI editing) coordinated via BullMQ/Redis queues, with job state in PostgreSQL and media on AWS S3 via presigned URLs.
- Built an AI media-editing microservice routing 9 edit types (object removal, background replacement, color grading, image-to-video) to Google Gemini through a unified routing layer with per-edit-type prompt templates.
- Developed a speech-to-caption microservice (FFmpeg + AssemblyAI) with word-level timed captions and a broadcast-standard segmenter enforcing reading-speed, line-length, and duration constraints.
- Added production resilience patterns — retries with backoff, concurrency limiting, idempotent job handling, and a retryable-vs-permanent error taxonomy — plus 12+ unit test suites with deterministic fixtures.

#### Project: Xelta SmartOps — Internal AI Operations & Observability Platform

- Designed a TypeScript monorepo (Turborepo, pnpm) delivering QA Guardian (automated credential audits across 13+ AI providers), XLP (a zero-loss ClickHouse/Redis Streams logging stack with a custom Node.js SDK), and Heartbeat (infra health monitoring built on AWS resources provisioned by the DevOps team).

#### Project: Payments & Credit Infrastructure (delivered to production)

- Engineered a multi-tenant Credit Management Microservice (NestJS, Prisma, PostgreSQL, Redis) implementing a Two-Phase Commit reservation pattern (RESERVE → COMMIT → RELEASE) to eliminate credit over-consumption across concurrent requests.
- Implemented atomic, idempotent credit deductions via row-level locking and unique jobId constraints, preventing double-deductions on retries with zero external locking infrastructure.
- Architected a multi-gateway payment service (Stripe, PayPal, Razorpay, Apple IAP, Google Play) processing subscriptions, webhooks, and PAYG credits via a BullMQ async worker pipeline with strict idempotency guarantees.
- Engineered fault-tolerant distributed workers using Redis TTL heartbeats and a Worker Guard decorator, eliminating ghost worker race conditions and guaranteeing exactly-once job execution.

### Backend Engineer — Node.js / Golang, Tourmates Pvt Ltd

Jul 2024 – Jan 2026

Backend Intern Jul–Oct 2024; converted to full-time — Gurugram, India

- Developed a high-performance Search Service in Golang with Elasticsearch and Redis, reducing average search latency by 40% via state-wise data sharding and fuzzy search handling.
- Engineered a soft-lock inventory reservation system using Redis Redlock and atomic Lua scripts to prevent double-bookings under high-concurrency requests, with TTL-based auto-release on payment abandonment.
- Implemented a Kafka-based event-driven pipeline to sync inventory state to Elasticsearch in near real-time, reducing indexing lag by over 90%.
- Established gRPC/Protobuf inter-service communication, replacing REST/JSON for internal calls to cut serialization overhead; deployed on AWS EC2 with Docker, PM2, and GitHub Actions CI/CD.
- Built an RBAC Admin Dashboard supporting HR, Operations, and Super Admin approval workflows with role-scoped access control.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Golang, Lua, SQL

**Backend:** Node.js, NestJS, Express.js, Fastify, REST APIs, gRPC, Microservices Architecture, System Design, Scalability, High Availability

**Databases & Caching:** PostgreSQL, MongoDB, Redis, ClickHouse, Prisma ORM, Elasticsearch, Redis Redlock, Fuzzy Search, Data Sharding

**Queue & Messaging:** BullMQ, Kafka, Redis Streams, Event-Driven Architecture, WebSockets, Socket.io

**Media & AI:** FFmpeg (GPU-aware encoding, scene detection), AssemblyAI, Google Gemini, proxy transcoding, Claude Code, Spec-Driven Development

**Payments & Security:** Stripe, PayPal, Razorpay, Apple IAP, Google Play Billing, JWT, RBAC, webhook verification, idempotency keys, PII redaction

**DevOps & Cloud:** Docker, Docker Compose, AWS (EC2, S3), PM2, GitHub Actions, CI/CD, Linux, Git, Turborepo, pnpm

**System Design Concepts:** Distributed Locking, Two-Phase Commit, Idempotency, Atomic Lua Scripting, Race Condition Prevention, Concurrency Control, Zero-Downtime Deployments

**Testing & Observability:** Jest, unit/integration testing, error fingerprinting, canary health checks, health-check endpoints, Bull Board

## EDUCATION

Full Stack Web Development — Coding Ninjas (Career Camp)

B.Tech, Mechanical Engineering — Sat Kabir Institute of Technology & Management

Diploma, Mechanical Engineering — DPG Institute of Technology & Management